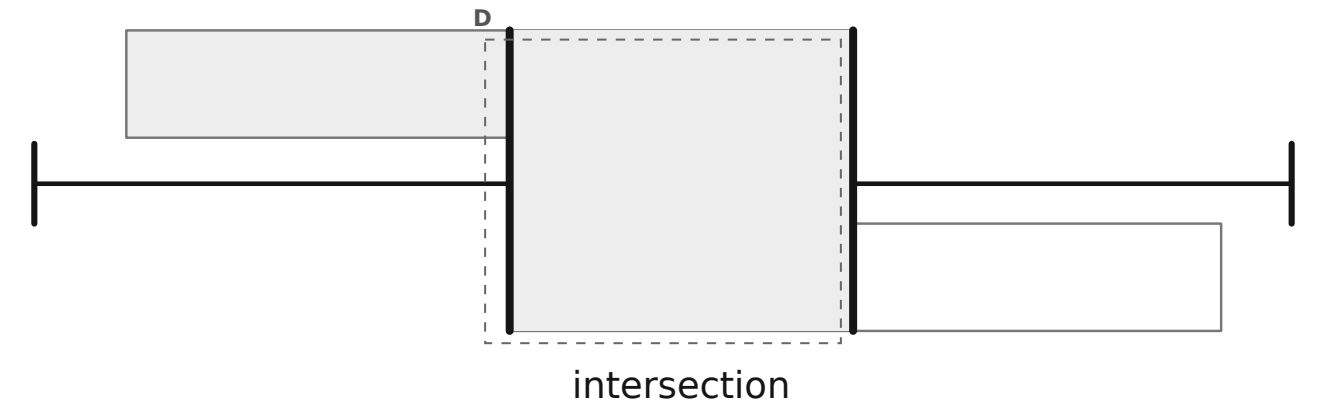
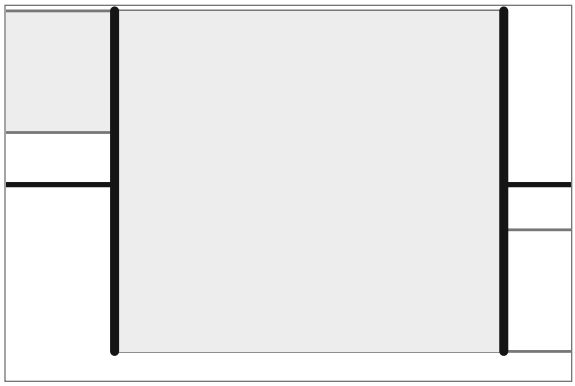
M040

Exact root bounds and coefficient feasibility



D / OPERATION DETAIL



Vector detail / 1.13× main-view scale

HOW THE OPERATION WORKS

Integer roots and inequality bounds determine which digit combinations could reconstruct an integer inside the advertised interval.

READ THE THREE STAGES

01 root bounds
02 feasible coefficients
03 reconstruct in range

WHAT WAS ACTUALLY DONE

Exact arithmetic helpers enumerate feasible roots, clip remainders and verify reconstruction. They reject impossible combinations, while still retaining every feasible unknown scalar unless a separate heuristic cuts further.

ORIGINAL FINDING / EXECUTED RECORD

Integer roots, coefficient limits, clipping and reconstruction are tested arithmetically.